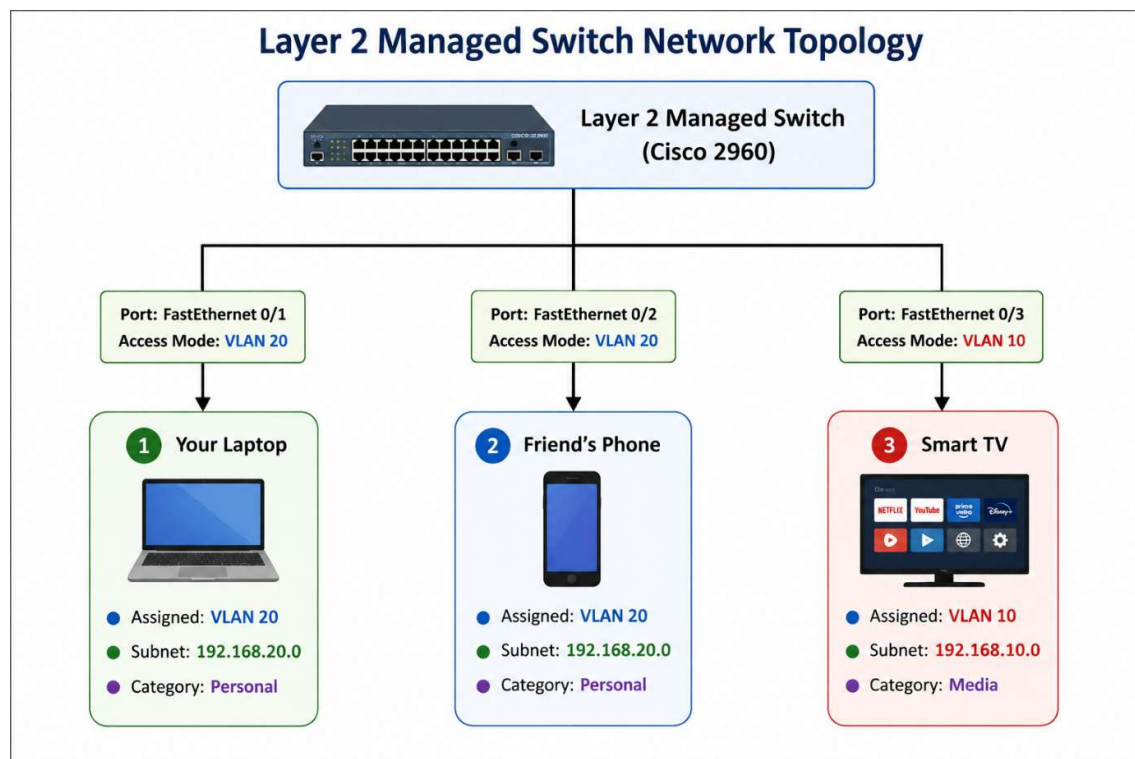


Task 1:

Multi-VLAN Switch Network Topology Diagram

Objective: Design a Layer 2 switch topology diagram segmenting personal devices and media streaming devices into separate Virtual Local Area Networks (VLANs).

1. Network Architecture Diagram:



2. Port & VLAN Assignment Table:

Connected Device	Switch Interface Port	Assigned VLAN ID	VLAN Name	Subnet Network	Traffic Isolation Purpose
Your Laptop	FastEthernet 0/1	VLAN 20	Personal_Dev	192.168.20.0/24	Isolated compute traffic
Friend's Phone	FastEthernet 0/2	VLAN 20	Personal_Dev	192.168.20.0/24	Allowed LAN access to Laptop
Smart TV	FastEthernet 0/3	VLAN 10	Streaming_Media	192.168.10.0/24	Blocked from

Connected Device	Switch Interface Port	Assigned VLAN ID	VLAN Name	Subnet Network	Traffic Isolation Purpose
					accessing Personal VLAN

3. Technical Summary:

- **VLAN 10 (Streaming):** Dedicated high-bandwidth media network. Devices in this VLAN cannot communicate with VLAN 20 at Layer 2 without a router (Inter-VLAN Routing).
- **VLAN 20 (Personal Devices):** Segregated user endpoints providing broadcast domain isolation and security against unauthorized network scanning.

Task 3:

Moving a PC from VLAN 10 to VLAN 20 & Verification

Objective: Demonstrate the CLI configuration steps to move a host interface (Port FastEthernet 0/1) from VLAN 10 to VLAN 20 and verify the resulting change in network connectivity.

1. Initial State (BEFORE Configuration):

- **PC1 Connected Interface:** Switch Port FastEthernet 0/1
- **Assigned VLAN:** VLAN 10 (Sales_VLAN)
- **Initial IP Address:** 192.168.10.10/24

CLI Verification Command (Initial State):

```
Switch# show mac address-table interface fastEthernet 0/1
```

```
Switch# show vlan brief
```

Output Before Move:

VLAN Name	Status	Ports
-----------	--------	-------

10	Sales_VLAN	active	Fa0/1, Fa0/2
20	Marketing_VLAN	active	Fa0/3

Initial Connectivity Test (From PC1):

C:\> ping 192.168.10.20 (PC2 in VLAN 10) --> SUCCESS
(Reply received)

C:\> ping 192.168.20.30 (PC3 in VLAN 20) --> FAILED
(Request timed out)

2. Switch CLI Reconfiguration Commands:

Execute the following commands on the switch to move
Port FastEthernet 0/1 into **VLAN 20**:

```
Switch> enable
```

```
Switch# configure terminal
```

```
Switch(config)# interface fastEthernet 0/1
```

```
Switch(config-if)# switchport mode access
```

```
Switch(config-if)# switchport access vlan 20
```

```
Switch(config-if)# exit
```

```
Switch(config)# exit
Switch# write memory
```

. Final State (AFTER Configuration):

- **PC1 Connected Interface:** Switch Port FastEthernet 0/1
- **New Assigned VLAN:** VLAN 20 (Marketing_VLAN)
- **Updated Host IP Address:** 192.168.20.10/24

CLI Verification Command (After Move):

```
Switch# show vlan brief
```

Output After Move:

VLAN Name	Status	Ports

10 Sales_VLAN	active	Fa0/2
20 Marketing_VLAN	active	Fa0/1, Fa0/3

Final Connectivity Test (From PC1):

C:\> ping 192.168.10.20 (PC2 in VLAN 10) --> FAILED
(Traffic blocked by VLAN isolation)

C:\> ping 192.168.20.30 (PC3 in VLAN 20) --> SUCCESS
(Reply from 192.168.20.30: bytes=32 time<1ms TTL=128)

Task 4:

Automating Bulk Port Assignment to VLAN 30 Using Interface Range Command

Objective: Automate the assignment of 10 switch ports (Ports 1 through 10) to VLAN 30 in Cisco Packet Tracer / GNS3 using the CLI range feature to eliminate repetitive commands.

1. Automated Cisco IOS Command Script:

**! Step 1: Access Privileged EXEC and Global
Configuration Mode**

enable

configure terminal

! Step 2: Create VLAN 30 if not already present

vlan 30

name Operations_VLAN

exit

! Step 3: Use interface range to select 10 ports simultaneously (Fa0/1 - Fa0/10)

interface range fastEthernet 0/1 - 10

! Step 4: Configure mode and assign all 10 ports to VLAN 30 in a single operation

switchport mode access

switchport access vlan 30

no shutdown

exit

! Step 5: Save Configuration

write memory

2. Verification Command & Expected Output:

Execute the following command on the switch to verify that all 10 ports are assigned to VLAN 30:

Switch# show vlan brief

Expected Output:

VLAN Name	Status	Ports

1 default	active	Fa0/11, Fa0/12, Fa0/13...
30 Operations_VLAN	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4, Fa0/5, Fa0/6, Fa0/7, Fa0/8, Fa0/9, Fa0/10

3. Technical Benefits:

- **Efficiency:** Applying configuration changes to 10 ports in 3 lines of CLI commands reduces time and human error.
- **Consistency:** Ensures all ports in the range receive identical port security and VLAN membership parameters simultaneously.